Interest Rate Announcements and Small-Firm Credit: A Fictional Event Study

A. Example and B. Sample

SAMPLE DOCUMENT FOR DEMONSTRATING READRAY. The study, data and results below are invented and are not real research findings.

Abstract We study how unexpected policy-rate announcements relate to credit spreads charged to small firms in a fictional economy. Using a panel of 1,200 invented loan contracts observed between 2015 and 2022, we estimate an event-study regression around 32 announcement dates. A one-percentage-point surprise increase is associated with a 0.41 percentage-point rise in small-firm spreads within one month, compared with 0.18 points for large firms. The association is weaker when banks hold more liquid assets. Because announcement timing may coincide with other news, we interpret the estimates as associations rather than clean causal effects.

1 Introduction

Central banks change interest rates to influence borrowing costs across the economy, but the pass-through may differ by borrower size. Small firms rely more on bank credit and have fewer outside options, so lenders may adjust their terms more quickly. This paper asks whether surprise policy announcements are followed by larger spread changes for small firms than for large firms, and whether bank liquidity dampens that difference.

Our contribution is a transparent event-study design applied to contract-level spreads. We build on the idea of a bank lending channel, in which monetary policy affects the supply of loans through bank balance sheets.

2 Data

The fictional dataset contains 1,200 loan contracts issued by 15 banks to 640 firms. Firms with fewer than 50 employees are classified as small. For each contract we observe the spread over the policy rate, maturity, collateral, and the issuing bank's ratio of liquid assets to total assets. Announcement surprises are measured as the change in a one-month market rate in a 30-minute window around each announcement.

Table 1 reports summary statistics. Small firms pay higher average spreads and borrow at shorter maturities.

Table 1: Summary statistics by firm size (fictional data).

Group	Contracts	Mean spread (pp)	Mean maturity (months)	Collateralised (%)
Small firms	730	3.12	18	71
Large firms	470	1.64	41	38
All	1200	2.54	27	58

3 Methodology

We estimate the following regression for contract i issued by bank b in month t:

$$\Delta s_{i,t} = \alpha + \beta S_t + \delta (S_t \times Small_i) + \theta (S_t \times Small_i \times Liq_b) + \gamma' X_{i,t} + \varepsilon_{i,t} \quad (1)$$

where Δs is the change in the loan spread relative to the bank's previous comparable contract, S is the announcement surprise in percentage points, Small is an indicator for small firms, Liq is the bank's liquid-asset ratio, and X contains maturity and collateral controls. The coefficient δ captures the additional spread response of small firms. Standard errors are clustered by bank. Identification relies on the assumption that, within the 30-minute window, the surprise is unrelated to other information that affects loan pricing.

4 Results

Table 2 presents the main estimates. The coefficient on the interaction term is positive and statistically significant: small-firm spreads rise by 0.41 percentage points after a one-point surprise ($\beta + \delta$), compared with 0.18 points for large firms. The triple interaction θ is negative, indicating that the small-firm premium shrinks when banks are more liquid.

Table 2: Event-study estimates of equation (1) (fictional; standard errors in parentheses).

Coefficient	Estimate	Std. error
β (surprise)	0.18	(0.05)
δ (surprise × small)	0.23	(0.07)
θ (surprise × small × liquidity)	-0.52	(0.21)
Observations	1200	

Figure 1 plots the average spread change in the months around announcements, separately for small and large firms.

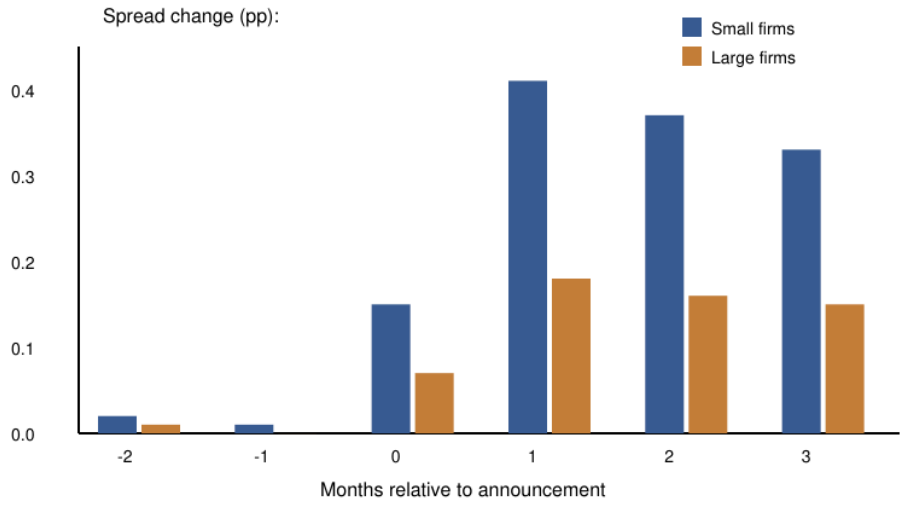


Figure 1: Average change in loan spreads (percentage points) by month relative to the announcement, small versus large firms (fictional data).

5 Limitations

Several limitations apply. First, other news released on announcement days could affect spreads, so the estimates may not be purely causal. Second, we observe only contracts that were approved; if banks ration credit to small firms instead of raising prices, spreads understate

the tightening. Third, the sample covers a single fictional country and period, which limits external validity.

6 Conclusion

In this fictional setting, surprise rate increases are associated with larger spread increases for small firms, and bank liquidity is associated with a smaller gap. Future work could examine loan volumes and rejection rates to capture credit rationing.

References

Example, A. (2020). A fictional review of the bank lending channel. *Journal of Illustrative Economics*, 1(1), 1–10.

Sample, B. (2021). Invented evidence on loan pricing. *Demonstration Finance Review*, 2(3), 11–20.